

MIND THE GAPS

MEASURING VISUAL ARTIFACTS IN DIMENSIONALITY REDUCTION

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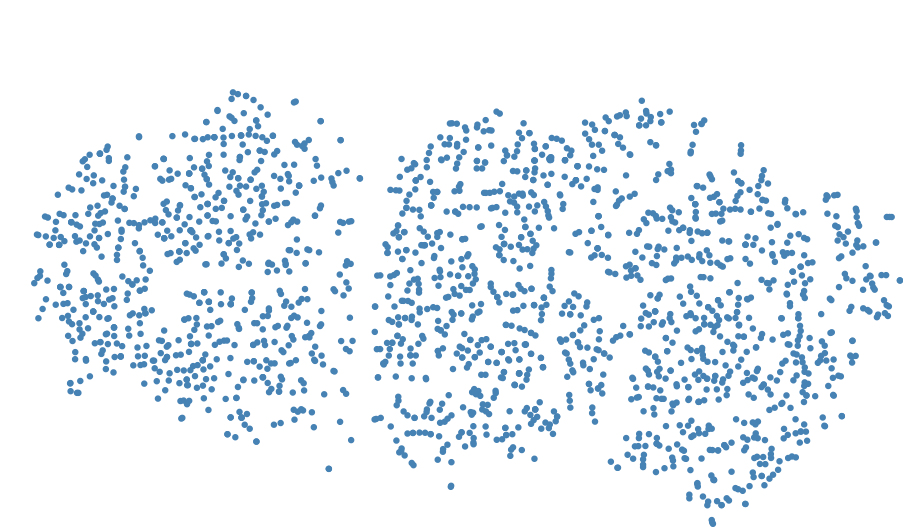
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Alessio Arleo

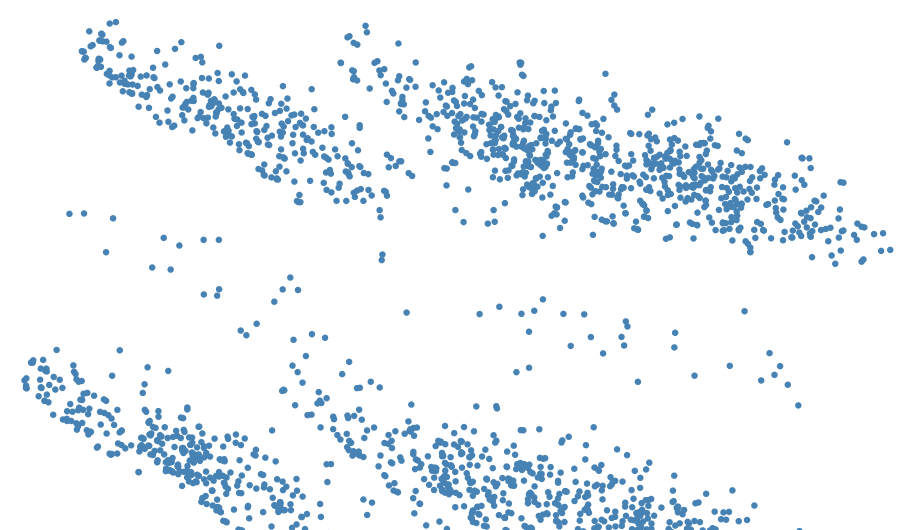
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Fernando Paulovich

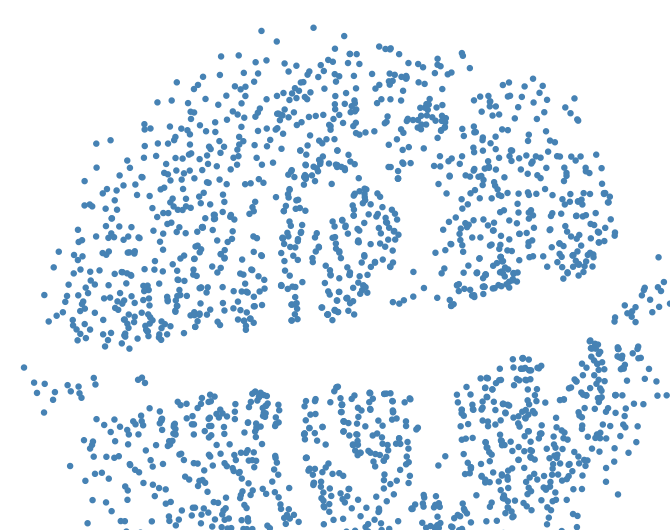
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Same dataset (*bank*), different DR methods. Which one to trust?

t-SNE



PCA



GFS [3]

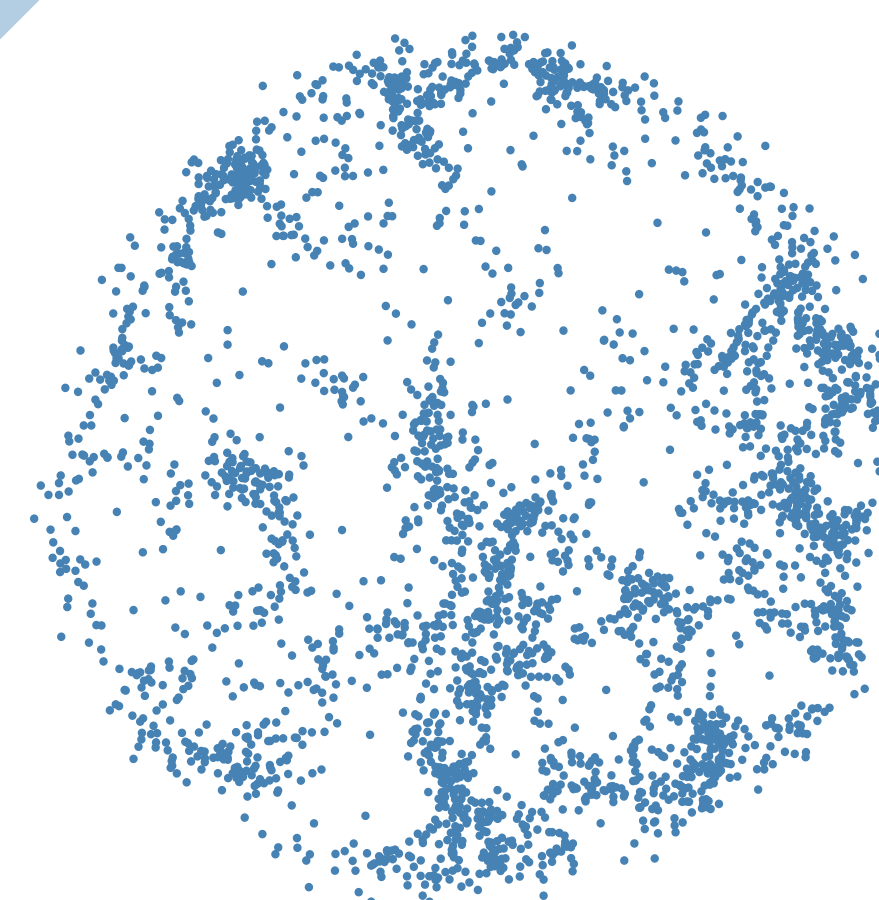
Many tasks rely on the **visual analysis** of scatterplots produced with dimensionality reduction.

Common *Projection Quality Metrics* capture the **structural preservation** of the data [1], but that does not always correspond with a correct **visual appearance** [2].

We know that these “gaps” are artifacts caused by a poor convergence of the DR method [3].

Common metrics such as stress and trustworthiness can struggle to **detect the presence of visual artifacts** that heavily **distort the visual aspect** of the layout.

The plot can have **empty regions**, or clusters of points, that **do not occur in the high-dimensional space**. Users may be **mislead about the structure** of the data.



FS

IMDB dataset

GFS

0.7187
0.5663
0.9574

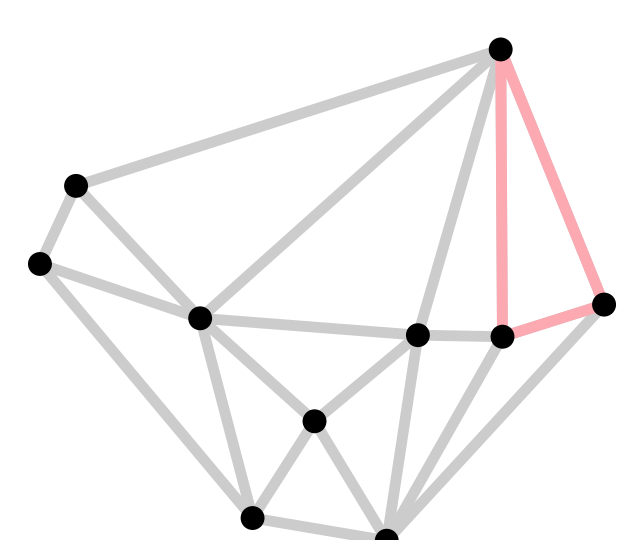
Stress (↓)
Trustworth. (↑)
Warp. Idx. (↓)

0.7307
0.5681
0.8966

WARPING INDEX

Measure if the space occupied by the gaps in the plot is correct.

1. Triangulate the projection



Each triangle $\hat{t}_i$ is composed of three points, that also define a triangle t_i in the original space [4].

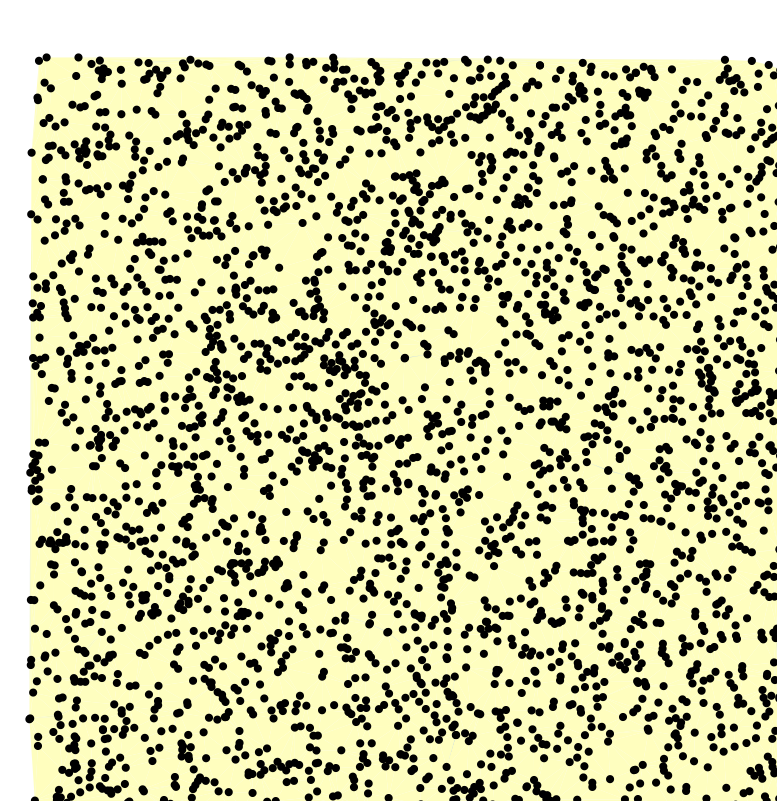
If a Delaunay triangulation is used, the WI has $O(n \log n)$ cost.

2. Quality of each triangle

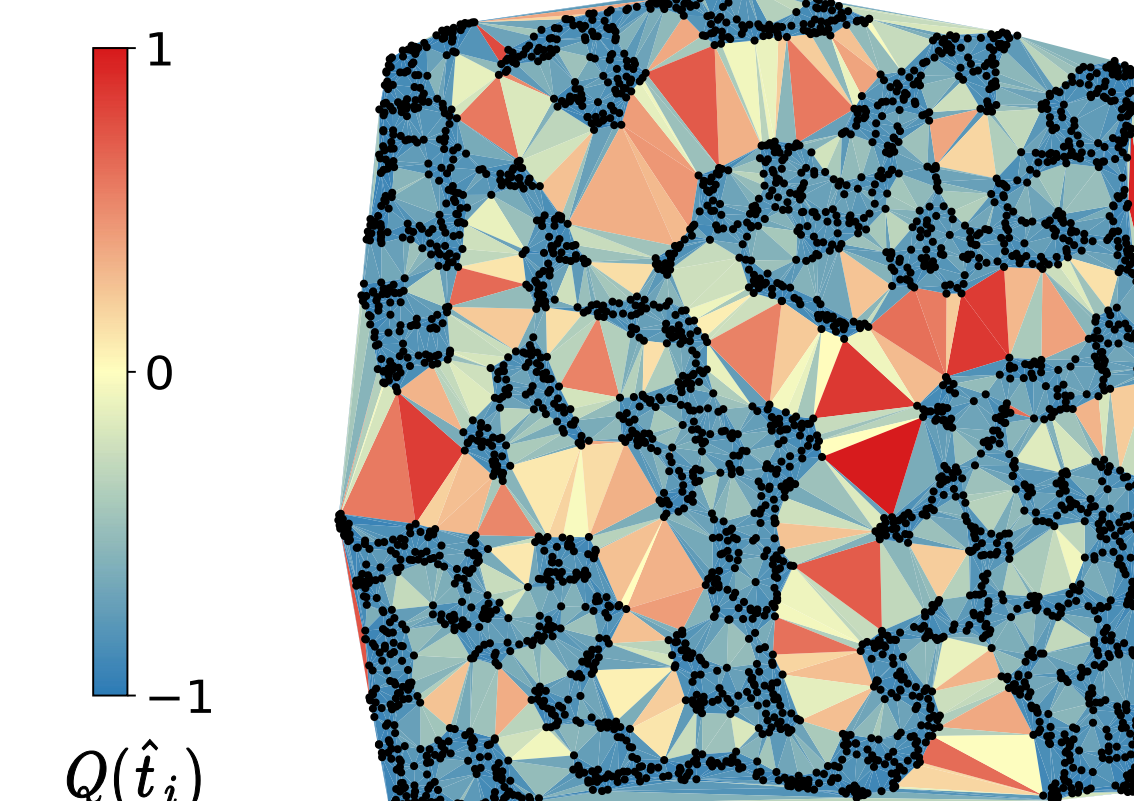
$$Q(\hat{t}_i) = \frac{A(\hat{t}_i) - A(t_i)}{\max(A(\hat{t}_i), A(t_i))} \in [-1, 1]$$

3. Warping Index

$$WI(P) = \frac{1}{\sum_{\hat{t}_i \in P} A(\hat{t}_i)} \sum_{\hat{t}_i \in P} A(\hat{t}_i) |Q(\hat{t}_i)| \in [0, 1]$$



PCA



t-SNE

0.0785
1
0.001

Stress (↓)
Trustworth. (↑)
Warp. Idx. (↓)

0.0397
1
0.7762

A 2D plane in 3D space is perfectly represented in 2D by PCA, while t-SNE introduces visual distortion.

Visualizing the triangle quality allows seeing where this distortion occurs.

The Warping Index quantifies the total distortion.

Which of these projections is less visually distorted?

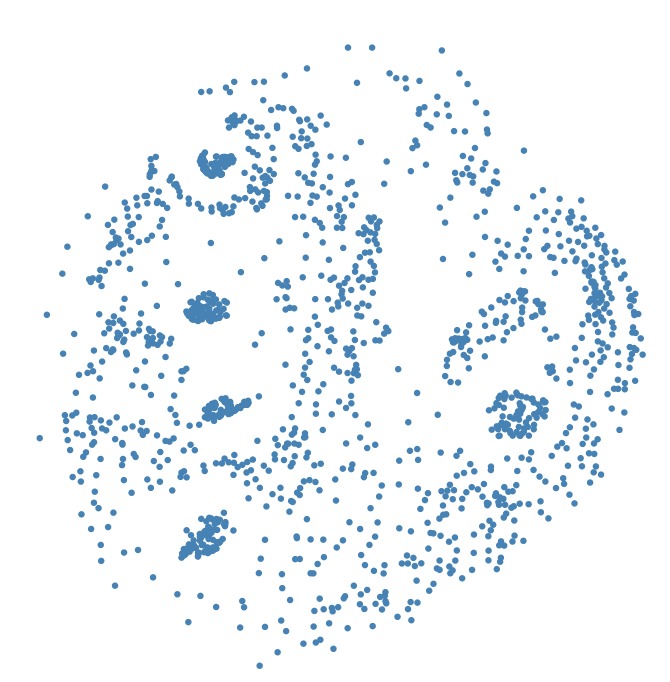
In this example, the Warping Index allows us to quantify that the GFS layout is the least distorted.



t-SNE



PCA



GFS

coil20 dataset

Stress (↓)
Trustworth. (↑)
Warp. Idx. (↓)

0.8834
0.9982
0.9537

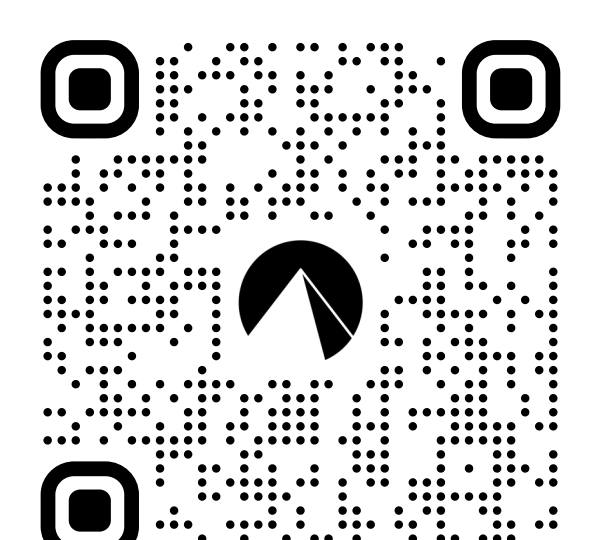
0.8797
0.8841
0.9463

0.8887
0.9128
0.7988

References

- [1] S. Johansson and J. Johansson. Interactive dimensionality reduction through user-defined combinations of quality metrics. *IEEE Transactions on Visualization and Computer Graphics*, 15(6):993–1000, 2009. doi: 10.1109/TVCG.2009.153
- [2] A. Machado, M. Behrisch, and A. Telea. Necessary but not sufficient: Limitations of projection quality metrics. *Computer Graphics Forum*, p. e70101, May 2025. doi: 10.1111/cgf.70101
- [3] J. Ros, A. Arleo, and F. V. Paulovich. Scalable Force Scheme: a fast method for projecting large datasets. In H.-J. Schulz and A. Vilanova, eds., *EuroVis Workshop on Visual Analytics (EuroVA)*. The Eurographics Association, 2025. doi: 10.2312/eurova.20251098
- [4] M. Aupetit. Visualizing distortions and recovering topology in continuous projection techniques. *Neurocomputing*, 70(7–9):1304–1330, Mar. 2007. doi: 10.1016/j.neucom.2006.11.018 2

Code and more


<https://codeberg.org/jros/warping-index>